

Deep Generative Models

Lecture 14

Roman Isachenko



AI Masters

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Recap of Previous Lecture

ELBO

$$\mathcal{L}_{\phi, \theta}(\mathbf{x}) = \mathbb{E}_{q(\mathbf{x}_1|\mathbf{x}_0)} \log p_{\theta}(\mathbf{x}_0|\mathbf{x}_1) - D_{\text{KL}}(q(\mathbf{x}_T|\mathbf{x}_0) \| p(\mathbf{x}_T)) - \underbrace{\sum_{t=2}^T \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} D_{\text{KL}}(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) \| p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t))}_{\mathcal{L}_t}$$

Discrete conditioned reverse distribution

$$q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) = \text{Cat}\left(\frac{\mathbf{Q}_t \mathbf{x}_t \odot \mathbf{Q}_{1:t-1} \mathbf{x}_0}{\mathbf{x}_t^{\top} \mathbf{Q}_{1:t} \mathbf{x}_0}\right).$$

- ▶ Both $q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)$ and $q(\mathbf{x}_t|\mathbf{x}_0)$ are known analytically from the forward process.
- ▶ The reverse process $p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t)$ is a learned categorical distribution:

$$p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t) = \text{Cat}(\boldsymbol{\pi}_{\theta}(\mathbf{x}_t, t)).$$

Recap of Previous Lecture

ELBO term

$$\mathcal{L}_t = \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} D_{\text{KL}}(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) \parallel p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t)).$$

Categorical KL

$$D_{\text{KL}}(\text{Cat}(\mathbf{q}) \parallel \text{Cat}(\mathbf{p})) = \sum_{k=1}^K q_k \log \frac{q_k}{p_k} = H(\mathbf{q}, \mathbf{p}) - H(\mathbf{q}),$$

- ▶ $H(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0))$ is a constant w.r.t. θ .
- ▶ $H(\mathbf{q}, \mathbf{p}) = -\sum_k q_k \log p_k$ is a **cross-entropy loss**.

Therefore, minimizing $\mathcal{L}_t$ w.r.t. θ is equivalent to minimizing

$$\mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} H(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0), p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t)).$$

Outline

1. Course Overview

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Course Overview: Problem Statement

Goal

Learn a generative model $p_{\theta}(\mathbf{x})$ that matches the data distribution $p_{\text{data}}(\mathbf{x})$.

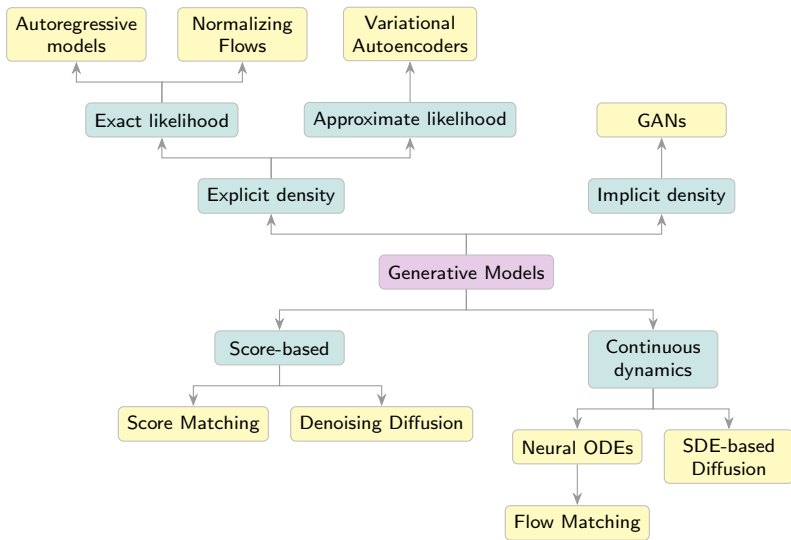
Three lenses on the same problem

- **Divergence minimization** (L1, L5):

$$\min_{\theta} D(p_{\text{data}} \parallel p_{\theta}) \quad (\text{KL, JS, Wasserstein, } \dots)$$

- **Likelihood-based** (L1–L4, L7–L8): maximize $\log p_{\theta}(\mathbf{x})$ (AR, NF, VAE, diffusion as hierarchical VAE).
- **Score-based** (L6–L10): learn $\nabla_{\mathbf{x}} \log p(\mathbf{x})$ (SM / NCSN, DDPM, score-based SDE).

Course Roadmap



Likelihood-based: Exact & Approximate Density (L1–L4)

Autoregressive (L1)

$$p_{\theta}(\mathbf{x}) = \prod_{i=1}^m p_{\theta}(x_i | \mathbf{x}_{1:i-1}) \quad \text{exact likelihood, sequential sampling}$$

Normalizing Flows (L2)

$$\mathbf{x} = \mathbf{f}_{\theta}(\mathbf{z}), \quad \log p_{\theta}(\mathbf{x}) = \log p(\mathbf{z}) + \log \left| \det \frac{\partial \mathbf{z}}{\partial \mathbf{x}} \right|$$

VAE / VQ-VAE (L3–L4)

$$\log p_{\theta}(\mathbf{x}) \geq \mathbb{E}_{q_{\phi}(\mathbf{z}|\mathbf{x})} \log p_{\theta}(\mathbf{x}|\mathbf{z}) - D_{\text{KL}}(q_{\phi}(\mathbf{z}|\mathbf{x}) \parallel p(\mathbf{z}))$$

- ▶ Amortized inference + reparametrization trick.
- ▶ ELBO surgery $\Rightarrow$ optimal prior; VQ-VAE with discrete latents.

Implicit & Score-based Foundations (L5–L6)

GAN / WGAN (L5)

- ▶ Implicit p_θ , adversarial min-max

$$\min_{\theta} \max_{\phi} \mathbb{E}_{p_{\text{data}}} \log D_{\phi}(\mathbf{x}) + \mathbb{E}_{p(\mathbf{z})} \log(1 - D_{\phi}(G_{\theta}(\mathbf{z})))$$

- ▶ Optimal discriminator $\Rightarrow$ JSD; WGAN uses Wasserstein distance via Kantorovich–Rubinstein duality.
- ▶ Evaluation tools: FID, Precision/Recall, CLIP score, human eval.

Score Matching / NCSN (L6)

- ▶ Learn $\mathbf{s}_{\theta}(\mathbf{x}) \approx \nabla_{\mathbf{x}} \log p_{\text{data}}(\mathbf{x})$ via denoising SM.
- ▶ Sample with (annealed) Langevin dynamics over multiple noise scales.

Diffusion Models (L7–L8)

Gaussian diffusion (L7)

- ▶ Forward $q(\mathbf{x}_t|\mathbf{x}_{t-1})$, direct sampling $q(\mathbf{x}_t|\mathbf{x}_0) = \mathcal{N}(\sqrt{\bar{\alpha}_t}\mathbf{x}_0, (1 - \bar{\alpha}_t)\mathbf{I})$.
- ▶ Diffusion as hierarchical VAE; ELBO splits into reconstruction + prior KL + $\sum_t \mathcal{L}_t$.

DDPM (L8)

$$\mathcal{L}_{\text{simple}} = \mathbb{E}_{t, \mathbf{x}_0, \epsilon} \left\| \epsilon - \epsilon_{\theta}(\mathbf{x}_t, t) \right\|^2, \quad \epsilon_{\theta} \propto -\sigma_t \mathbf{s}_{\theta}$$

- ▶ Equivalent to score-based generative modeling (NCSN-style).

Guidance (L8)

$$\hat{\epsilon}_{\theta}(\mathbf{x}_t, t, \mathbf{y}) = (1 - \gamma) \epsilon_{\theta}(\mathbf{x}_t, t, \emptyset) + \gamma \epsilon_{\theta}(\mathbf{x}_t, t, \mathbf{y})$$

Continuous Dynamics & SDE (L9–L10)

Neural ODE / continuous-time NF (L9)

$$\frac{d\mathbf{x}(t)}{dt} = \mathbf{v}_{\theta}(\mathbf{x}(t), t), \quad \frac{d}{dt} \log p_t(\mathbf{x}(t)) = -\text{tr}\left(\frac{\partial \mathbf{v}_{\theta}}{\partial \mathbf{x}}\right)$$

Forward SDE (L9)

$$d\mathbf{x} = \mathbf{f}(\mathbf{x}, t) dt + g(t) d\mathbf{w}$$

- KFP equation; VE / VP SDEs unify NCSN and DDPM.

Reverse SDE & Probability flow ODE (L10)

$$d\mathbf{x} = [\mathbf{f}(\mathbf{x}, t) - g^2(t) \mathbf{s}_{\theta}(\mathbf{x}, t)] dt + g(t) d\bar{\mathbf{w}}$$
$$\frac{d\mathbf{x}}{dt} = \mathbf{f}(\mathbf{x}, t) - \frac{1}{2}g^2(t) \mathbf{s}_{\theta}(\mathbf{x}, t) \quad (\text{deterministic equivalent})$$

Flow Matching (L10–L11)

Flow Matching objective (L10)

$$\mathcal{L}_{\text{FM}} = \mathbb{E}_{t, p_t(\mathbf{x})} \left\| \mathbf{v}_\theta(\mathbf{x}, t) - \mathbf{v}(\mathbf{x}, t) \right\|^2$$

Target $\mathbf{v}(\mathbf{x}, t)$ is the marginal vector field — intractable in general.

Conditional Flow Matching (L11)

$$\mathcal{L}_{\text{CFM}} = \mathbb{E}_{t, p(\mathbf{z}), p_t(\mathbf{x}|\mathbf{z})} \left\| \mathbf{v}_\theta(\mathbf{x}, t) - \mathbf{v}(\mathbf{x}, \mathbf{z}, t) \right\|^2$$

- ▶ $\nabla_\theta \mathcal{L}_{\text{FM}} = \nabla_\theta \mathcal{L}_{\text{CFM}}$ — regress against *conditional* field.
- ▶ Conditioning on endpoint $\mathbf{z} = \mathbf{x}_1$ (one-sided) or pair $(\mathbf{x}_0, \mathbf{x}_1)$ (two-sided).
- ▶ Straight Gaussian paths $\Rightarrow$ closed-form $\mathbf{v}(\mathbf{x}, \mathbf{z}, t) = \mathbf{x}_1 - \mathbf{x}_0$.

Discrete Diffusion (L12–L13)

Forward discrete process (L12)

Categorical $\mathbf{x}_t$ with transition matrices $\mathbf{Q}_t$:

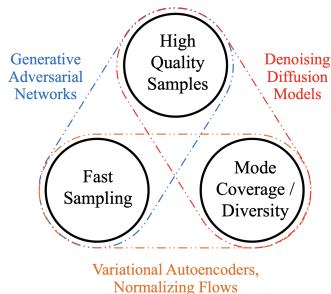
- ▶ **Uniform:** $\mathbf{Q}_t = (1 - \beta_t)\mathbf{I} + \beta_t\mathbf{U}$
- ▶ **Absorbing:** $\mathbf{Q}_t = (1 - \beta_t)\mathbf{I} + \beta_t\mathbf{e}_m\mathbf{1}^\top$ (mask token)

Reverse process & Discrete ELBO (L13)

$$q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) = \text{Cat}\left(\frac{\mathbf{Q}_t\mathbf{x}_t \odot \mathbf{Q}_{1:t-1}\mathbf{x}_0}{\mathbf{x}_t^\top \mathbf{Q}_{1:t}\mathbf{x}_0}\right),$$

each $\mathcal{L}_t$ reduces to a **cross-entropy** loss against $p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)$.

Generative Learning Trilemma



Rule of thumb

- ▶ **Likelihood + Coverage** $\Rightarrow$ **AR / NF**
exact density, *slow sampling*.
- ▶ **Likelihood + Fast Sampling** $\Rightarrow$ **VAE**
tractable bound, *blurry samples*.
- ▶ **Quality + Fast Sampling** $\Rightarrow$ **GAN**
sharp, *mode collapse, no likelihood*.
- ▶ **Quality + Coverage** $\Rightarrow$ **Diffusion**
stable training, *slow sampling*.
- ▶ **Quality + Faster Sampling** $\Rightarrow$ **FM / ODE**
fewer steps, *approx. likelihood*.
- ▶ **Discrete + Coverage** $\Rightarrow$ **Discrete diffusion**
parallel denoising, *iterative decoding*.

Summary

- ▶ No generative model dominates all axes of the trilemma — each family trades off likelihood, sample quality, mode coverage and sampling speed.
- ▶ Modern frontier (flow matching, discrete diffusion, distilled diffusion) tries to push the Pareto frontier of this trilemma.
- ▶ Key takeaway: pick a method by the dominant constraint of your application — likelihood vs. sample quality vs. latency vs. data modality.